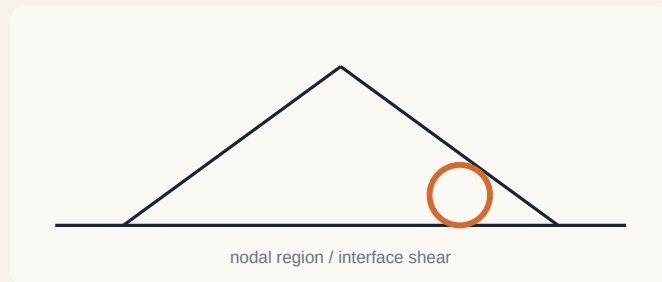


Temporary Is Still a Load Case

Issue 002 · original cases · reconstructed 18 Aug 2026

DEEP DIVE · FIU PEDESTRIAN BRIDGE



YOU ARE THE ENGINEER

A 174 ft concrete pedestrian-bridge span has been erected across an eight-lane highway. Large cracks appear around a heavily loaded truss-node region and continue growing for 19 days. A remedial plan proposes retensioning post-tensioning rods while traffic remains below. Before thinking about the repair, what evidence would you require before allowing anyone above, below or beside the structure?

On 15 March 2018, while workers were retensioning rods in truss member 11, concrete failed near member 12 and the truss lost geometric stability. The span fell about 18.5 ft onto SW 8th Street, killing six people.

NTSB determined that design calculation errors in the member 11/12 nodal region and its deck connection were the probable cause. Actual demand was nearly twice what the designers had calculated, while shear capacity had also been overestimated.

THE SECOND FAILURE: CHECKING

The independent peer review did not check the unusual truss nodes or all construction stages. The reviewing firm was also found not to be appropriately qualified for the task. A peer review is not a ceremonial stamp: it only protects you if the unusual parts are actually recalculated.

ENGINEER'S NOTEBOOK

When cracking is dramatically beyond what your model predicts, do not immediately design a repair using the same model that failed to predict the distress. First establish why the structure is disagreeing with you.

SOURCES FOR THIS CASE

NTSB · Pedestrian Bridge Collapse Over SW 8th Street, Miami · HAR-19/02
<https://www.nts.gov/investigations/AccidentReports/Reports/HAR1902.pdf>

Problems from Practice

Construction-stage buoyancy + fixing installation

SITE PROBLEM · THE UNDERGROUND TANK BECAME A SUBMARINE

uplift > construction-stage dead weight



A reinforced-concrete underground tank near Mangalore measured about 90 m × 35 m internally and 7.3 m deep, with a 1.0 m raft. The side walls were complete and the top slab was still under construction when the monsoon arrived. The middle of the raft rose roughly 300 mm and an internal partition wall cracked.

For the completed tank, calculated uplift factor of safety had been about 1.39. At the actual construction stage, dead load was lower; then groundwater rose about 3.4 m above the design level. The reported construction-stage factor of safety fell to about **0.67**.

THE HYDROGEOLOGY WAS THE STORY

Post-failure investigation found saturated strata, loose pockets and subsurface flow paths. An old perennial well had been filled during construction and covered by a slab; local geology and topography helped develop excessive hydraulic head beneath the raft.

WHAT THEY DID

With drilling access prevented by heavy rain, engineers diverted water into the tank itself, adding downward weight until it settled back. Permanent works then used twelve 500 mm diameter, 20 m deep dewatering wells with automatic submersible pumps. No manufacturer/model was reported.

SOURCE

ISSMGE · Uplift of an Underground Tank in Northern Malabar Region, India
<https://geocasehistoriesjournal.org/issues/volume-4-2/uplift-of-an-underground-tank-in-northern-malabar-region-india/pdf>

DETAIL / FIXING · INSTALLATION IS PART OF STRUCTURAL CAPACITY



installation procedure is part of capacity

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QA LESSON

The contractor introduced paint marking to prevent inadvertent reuse and pull testing to verify capacity. CROSS's panel noted that sacrificial testing may be needed where correct installation cannot be independently verified and advised against relying on a single fixing in safety-critical situations.

SPECIFICATION LESSON

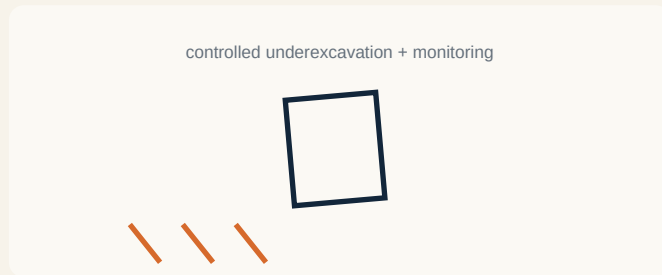
A proprietary concrete/masonry anchor screw failed in a handrail application and a post fell from height. CROSS reported that the manufacturer's installation instructions had not been followed.

Those instructions covered hole diameter, hole depth, cleaning, installation method, maximum tightening torque, avoiding hammering, and single-use requirements. These are not administrative niceties. They are part of the tested system that produces the published resistance.

Engineering Done Well

Pisa stabilisation · issue synthesis

ENGINEERING WIN · CONTROLLED UNDEREXCAVATION AT PISA



By 1990, the Leaning Tower of Pisa had rotated far enough south that the Italian government appointed an international multidisciplinary committee and closed the monument. Foundation pressure was extremely eccentric and the problem was not simply conventional bearing-capacity failure.

Long-term measurements showed the north side rising while the south side descended, consistent with a leaning-instability mechanism. Seasonal groundwater behaviour also correlated with small southward rotations.

Engineers first bought time with roughly 600 tonnes of lead counterweight on the north side, later increasing it during an unsuccessful ground-anchor attempt. For the principal correction they selected **underexcavation**: carefully removing tiny volumes of soil beneath the high north side so the tower would rotate back toward vertical.

OBSERVATIONAL METHOD WITH TEETH

A specially developed drilling system extracted soil with minimal disturbance. Each day's measured response was reviewed by the responsible engineer, who then issued signed instructions specifying the next day's extraction location and volume.

Full underexcavation used 41 holes at about 0.5 m spacing north of the tower. By June 2001, the tower had rotated northward roughly 1800 arc-seconds and its inclination had been reduced by about 10% relative to the 1993 maximum.

ENGINEER'S NOTEBOOK

Monitoring becomes design only when measurements are allowed to change the next action. The cycle is: extract a little → measure → compare → decide → repeat.

SOURCE

ISSMGE / Burland et al. · **Stabilising the Leaning Tower of Pisa**
https://www.issmge.org/uploads/publications/1/44/2009_020.pdf

THE THREAD · TEMPORARY STATES ARE STRUCTURAL STATES

FIU: construction stages and unusual nodes were not adequately checked. **Underground tank**: the completed structure had enough dead weight, while the half-built structure under monsoon groundwater did not. **Anchor screw**: structural resistance depended on installation procedure. **Pisa**: temporary counterweights and a staged extraction sequence were central structural interventions, not logistics.

60-SECOND TAKEAWAY · Always ask what the least-stable configuration is between “nothing exists” and “finished structure.” Check buoyancy at the lightest construction stage. Treat fixing installation as part of capacity. Monitoring is most valuable when it changes the next engineering decision.